

- 7 (a) X-rays are produced in an X-ray tube when high-speed electrons are accelerated toward and hit a metal target. Fig. 7.1 shows the variation with wavelength of the intensity of X-ray radiation emitted.

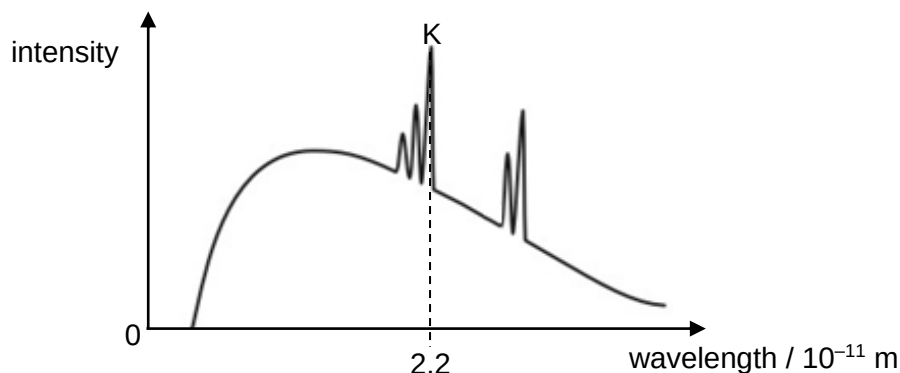


Fig. 7.1

- (i) Explain why there is a continuous distribution of wavelengths.

X-ray photons produced when high-speed bombarding electrons lose kinetic energy when they collide with target metal atoms OR decelerate OR X-ray radiation produced when electrons accelerated [B1] – explain process of how X-ray photons are produced

Electrons lose a range/distribution of kinetic energies OR range/distribution of decelerations OR range/distribution of accelerations, so range/distribution of X-ray wavelengths emitted [B1] – explain why continuous wavelengths [2]

- (ii) A series of characteristic lines shown by the high intensity peaks, such as K, are observed in Fig. 7.1.

1. Calculate the energy difference, in keV, associated with the characteristic line K.

$$\begin{aligned}\Delta E &= \frac{hc}{\lambda} \\ &= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2.2 \times 10^{-11}} \quad [\text{C1}] \\ &= 9.04 \times 10^{-15} \text{ J} \\ \frac{9.04 \times 10^{-15}}{1.6 \times 10^{-19} \times 10^3} &= 56.5 \text{ keV} \quad [\text{A1}]\end{aligned}$$

energy difference = keV [2]

2. Suggest why there are other series of characteristic lines produced at wavelengths longer than K.

Collision with high-speed bombarding electron causes inner shell electron to be

knocked out, creating a vacancy refilled by outer shell electron. [B1] – describes the process

of producing characteristic lines

Vacancy are in shells above K-shell, X-ray photons emitted when these vacancies

are filled have smaller energy difference than K-series, hence longer wavelengths

[2]

[B1] – explain why longer wavelengths

Question 7 continues on the following page.

- (b) A simple model of an atom with one electron can be represented by the electron as a stationary wave confined in a box of length 1.0×10^{-10} m equal to the diameter of the atom, as shown in Fig. 7.2.

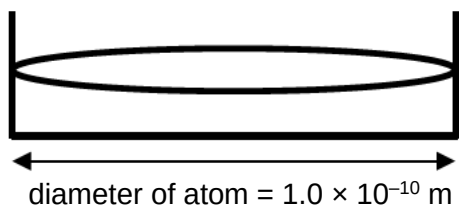


Fig. 7.2

- (i) State the uncertainty in locating the position of the electron.

uncertainty in position = 1.0×10^{-10} [A1] m [1]
OR 5.0×10^{-11}

- (ii) Calculate the uncertainty in the velocity of the electron.

$$\Delta x \Delta p \geq h$$

$$m \Delta v \geq \frac{h}{\Delta x}$$

$$\Delta v \geq \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(1.0 \times 10^{-10})}$$

$$\Delta v \geq 7.3 \times 10^6 \text{ m s}^{-1} \quad [\text{B1}]$$

uncertainty in velocity = m s⁻¹ [1]

- (iii) The diameter of a nucleus is 10^4 times smaller than the diameter of an atom.
Using the model above, suggest why an electron cannot be found inside the nucleus.

The uncertainty in the electron's velocity would be 10^4 times more ($7.3 \times 10^{10} \text{ m s}^{-1}$)

exceeding the speed of light, hence not possible to find an electron inside the nucleus [B1]

.....[1]

[Total: 9]

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Section B

Answer **one** question from this Section in the spaces provided.

2 This question is on the common lost topic of Nuclear Physics and will not be assessed